A SEPARATED FLAT LOCALLY FINITE TYPE MORPHISM WHICH IS AN ISOMORPHISM OVER A RETROCOMPACT SCHEME THEORETICALLY DENSE OPEN IS AN OPEN IMMERSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Finiteness of presentation over a scheme theoretically dense open, openness of a flat and locally finitely presented morphism, and dimension zero fibres. Same chapter flat.tex, labels lemma-flat-finite-type-finitely-presented-over-dense-open-X (proof line 3084) and lemma-flat-finite-presentation-dimension-over-dense-open (proof line 3094); Morphisms, label morphisms-lemma-fppf-open (proof lines 3087 and 3104). The first upgrades local finite type to local finite presentation, the second shows all fibres have dimension zero whence the morphism is quasi-finite, and openness of the image lets the base be replaced by the image of the morphism.
- The etale splitting of the quasi-finite part over an elementary etale neighbourhood. More on Morphisms, label more-morphisms-lemma-etale-splits-off-quasi-finite-part, invoked at proof line 3101; Morphisms, label morphisms-lemma-locally-quasi-finite-rel-dimension-0 (proof line 3096). Produces the elementary etale neighbourhood over which the base change splits into a finite part V and a part W empty over the closed fibre, which is the key representation of the morphism.
- Scheme theoretic density under flat base change, descent of finiteness, and finite flat morphisms being finite locally free. Morphisms, labels morphisms-lemma-flat-morphism-scheme-theoretically-dense-open (proof line 3110), morphisms-lemma-finite-flat (proof line 3118) and morphisms-lemma-finite-locally-free (proof line 3121); Descent, label descent-lemma-descending-property-finite (proof line 3116). Shows the inverse image of the dense open remains

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scheme theoretically dense so that the empty part W must vanish, lets finiteness descend back to the base, and upgrades finite flat to finite locally free so that the degree can be compared over the dense open.

Source reference. Original lemma 02LA: sources/stacks-chow/tags.txt, source lines 3401–3401. Navigation pointer for \ref{descent-lemma-descending-property-finite} used inside the counted span (line 3116); the argument that consumes it is retained in the span.

Source reference. Original lemma 081L: sources/stacks-flat/pinned-original-source.tex, source lines 3020–3020. Navigation pointer for \ref{lemma-flat-finite-presentation-dimension-over-dense-open} used inside the counted span (line 3094); the argument that consumes it is retained in the span.

Source reference. Original lemma 081P: sources/stacks-flat/pinned-original-source.tex, source lines 3000–3000. Navigation pointer for \ref{lemma-flat-finite-type-finitely-presented-over-dense-open-X} used inside the counted span (line 3084); the argument that consumes it is retained in the span.

Source reference. Original lemma 02LP: sources/stacks-chow/tags.txt, source lines 3416–3416. Navigation pointer for \ref{more-morphisms-lemma-etale-splits-off-quasi-finite-part} used inside the counted span (line 3101); the argument that consumes it is retained in the span.

Source reference. Original lemma 02KB: sources/stacks-chow/tags.txt, source lines 3366–3366. Navigation pointer for \ref{morphisms-lemma-finite-flat} used inside the counted span (line 3118); the argument that consumes it is retained in the span.

Source reference. Original lemma 04MH: sources/stacks-chow/tags.txt, source lines 6043–6043. Navigation pointer for \ref{morphisms-lemma-finite-locally-free} used inside the counted span (line 3121); the argument that consumes it is retained in the span.

Source reference. Original lemma 081H: sources/stacks-chow/tags.txt, source lines 10217–10217. Navigation pointer for \ref{morphisms-lemma-flat-morphism-scheme-theoretically-dense-open} used inside the counted span (line 3110); the argument that consumes it is retained in the span.

Source reference. Original lemma 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Navigation pointer for \ref{morphisms-lemma-fppf-open} used inside the counted span (line 3087); the argument that consumes it is retained in the span.

Source reference. Original lemma 0397: sources/stacks-chow/tags.txt, source lines 4262–4262. Navigation pointer for \ref{morphisms-lemma-locally-quasi-finite-rel-dimension-0} used inside the counted span (line 3096); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes and $U \subset S$ an open. If*

(1) *f is separated, locally of finite type, and flat,*
(2) *$f^{-1}(U) \to U$ is an isomorphism, and*
(3) *$U \subset S$ is retrocompact and scheme theoretically dense,*

then f is an open immersion.

Proof. By Lemma 081P the morphism f is locally of finite presentation. The image $f(X) \subset S$ is open (Morphisms, Lemma 01UA) hence we may replace S by $f(X)$. Thus we have to prove that f is an isomorphism. We may assume S is affine. We can reduce to the case that X is quasi-compact because it suffices to show that any quasi-compact open $X' \subset X$ whose image is S maps isomorphically to S. Thus we

may assume f is quasi-compact. All the fibers of f have dimension 0, see Lemma 081L. Hence f is quasi-finite, see Morphisms, Lemma 0397. Let $s \in S$. Choose an elementary étale neighbourhood $g : (T, t) \to (S, s)$ such that $X \times_S T = V \amalg W$ with $V \to T$ finite and $W_t = \emptyset$, see More on Morphisms, Lemma 02LP. Denote $\pi : V \amalg W \to T$ the given morphism. Since π is flat and locally of finite presentation, we see that $\pi(V)$ is open in T (Morphisms, Lemma 01UA). After shrinking T we may assume that $T = \pi(V)$. Since f is an isomorphism over U we see that π is an isomorphism over $g^{-1}U$. Since $\pi(V) = T$ this implies that $\pi^{-1}g^{-1}U$ is contained in V. By Morphisms, Lemma 081H we see that $\pi^{-1}g^{-1}U \subset V \amalg W$ is scheme theoretically dense. Hence we deduce that $W = \emptyset$. Thus $X \times_S T = V$ is finite over T. This implies that f is finite (after replacing S by an open neighbourhood of s), for example by Descent, Lemma 02LA. Then f is finite locally free (Morphisms, Lemma 02KB) and after shrinking S to a smaller open neighbourhood of s we see that f is finite locally free of some degree d (Morphisms, Lemma 04MH). But $d = 1$ as is clear from the fact that the degree is 1 over the dense open U. Hence f is an isomorphism. □